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1. $\lambda$ is an eigenvalue of a matrix A if and only if the equation A - $\lambda$x = 0 has a:

   a) Non-trivial solution (Correct option)

   b) Trivial solution

2. Which one of the following statements is true for all real symmetric matrices?

   a) All the eigenvalues are distinct

   b) Sum of all the eigenvalues is zero.

   c) All the eigenvalues are real. (Correct option)

   d) All the eigenvalues are positive.

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3. A matrix A is diagonalizable if and only if there are enough eigenvectors to form a basis of:

   a) None of the above (Correct option)

   b) $\mathbb{R}^3$

   c) $\mathbb{R}^{6n}$

   d) $\mathbb{R}^2$

4. A transformation is a rule that assigns to each vector x in _____, an image vector T(x) in _____.

   a) Domain of T, co-domain of T (Correct option)

   b) Co-domain of T, domain of T

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5. The dominant or principal eigenvector of a matrix is an eigenvector corresponding to the eigenvalue of _____ magnitude (for real numbers, largest absolute value) of that matrix.

a) All of these

b) Smallest

c) Zero

d) Largest (Correct option)

6. If x + 2 is a factor of the characteristic polynomial of matrix C, then an eigenvalue of C is:

a) -2 (Correct option)

b) 2

c) 0

d) 1/2

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7. An n x n matrix with n distinct eigenvalues is:

a) Diagonalizable (Correct option)

b) Hermitian

c) Invertible

d) Symmetric

8. Let A be an n x n matrix whose distinct eigenvalues are $\lambda_1, \lambda_2, ..., \lambda_n$. For 1 $\leq$ k $\leq$ p, the dimension of the eigenspace for $\lambda_k$ is _____ than or _____ to the multiplicity of the eigenvalue.

a) Greater, equal (Correct option)

b) Less, equal

9. If A is a triangular matrix, then the eigenvalues of A are the entries on the main diagonal of A.

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a) False (Correct option)

b) True

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